

Module4 lab2

Корначенко Данил Александрович

1) На L2-01

```
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vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree vlan 1 root primary
vIOS-L2-01(config)#
```

2) Проверка доступности компов

The screenshot displays a network simulator interface with five PC windows (PC1 to PC6). Each window shows configuration details (IP, MASK, GATEWAY, DNS, MAC, LPORT, RHOST:PORT, MTU) and the results of a ping command to 192.168.1.4. Red arrows indicate connectivity from PC1 to PC2 and PC3.

PC	IP	MASK	GATEWAY	DNS	MAC	LPORT	RHOST:PORT	MTU	Ping Results (to 192.168.1.4)
PC1	192.168.1.1	24	192.168.1.254		08:00:27:00:00:00	20412	127.0.0.1:20413	1500	84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=8.112 ms 84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=15.886 ms 84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=29.666 ms 84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=10.821 ms 84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=29.037 ms
PC2	192.168.1.2	24	192.168.1.254		08:00:27:00:00:01	20414	127.0.0.1:20415	1500	84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=8.596 ms 84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=0.151 ms 84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=5.210 ms 84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=3.015 ms 84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=7.748 ms
PC3	192.168.1.3	24	192.168.1.254		08:00:27:00:00:02	20416	127.0.0.1:20417	1500	84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=29.361 ms 84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=8.856 ms 84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=5.894 ms 84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=14.228 ms 84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=11.288 ms
PC4	192.168.1.4	24	192.168.1.254		08:00:27:00:00:03	20418	127.0.0.1:20419	1500	
PC5	192.168.1.5	24	192.168.1.254		08:00:27:00:00:04	20420	127.0.0.1:20421	1500	84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=10.744 ms 84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=7.037 ms 84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=13.486 ms 84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=12.566 ms 84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=13.881 ms
PC6	192.168.1.6	24	192.168.1.254		08:00:27:00:00:05	20422	127.0.0.1:20423	1500	

Layer2Switch-1

```
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*****
VIOS-L2-01>
VIOS-L2-01>
VIOS-L2-01>
VIOS-L2-01>show spanning-tree

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 24577
Address 0c62.e6ee.0000
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24577 (priority 24576 sys-id-ext 1)
Address 0c62.e6ee.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec
```

Layer2Switch-3

```
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*****
VIOS-L2-01>show spanning-tree

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 24577
Address 0c62.e6ee.0000
Cost 4
Port 1 (GigabitEthernet0/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0ca2.81bf.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec
```

Layer2Switch-4

```
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*****
VIOS-L2-01>show spanning-tree

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 24577
Address 0c62.e6ee.0000
Cost 4
Port 1 (GigabitEthernet0/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0ca2.bbae.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec
```

Layer2Switch-2

```
VIOS-L2-01>show spanning-tree

VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 24577
Address 0c62.e6ee.0000
Cost 4
Port 1 (GigabitEthernet0/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0cb3.cd2e.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec
```

Layer2Switch-5

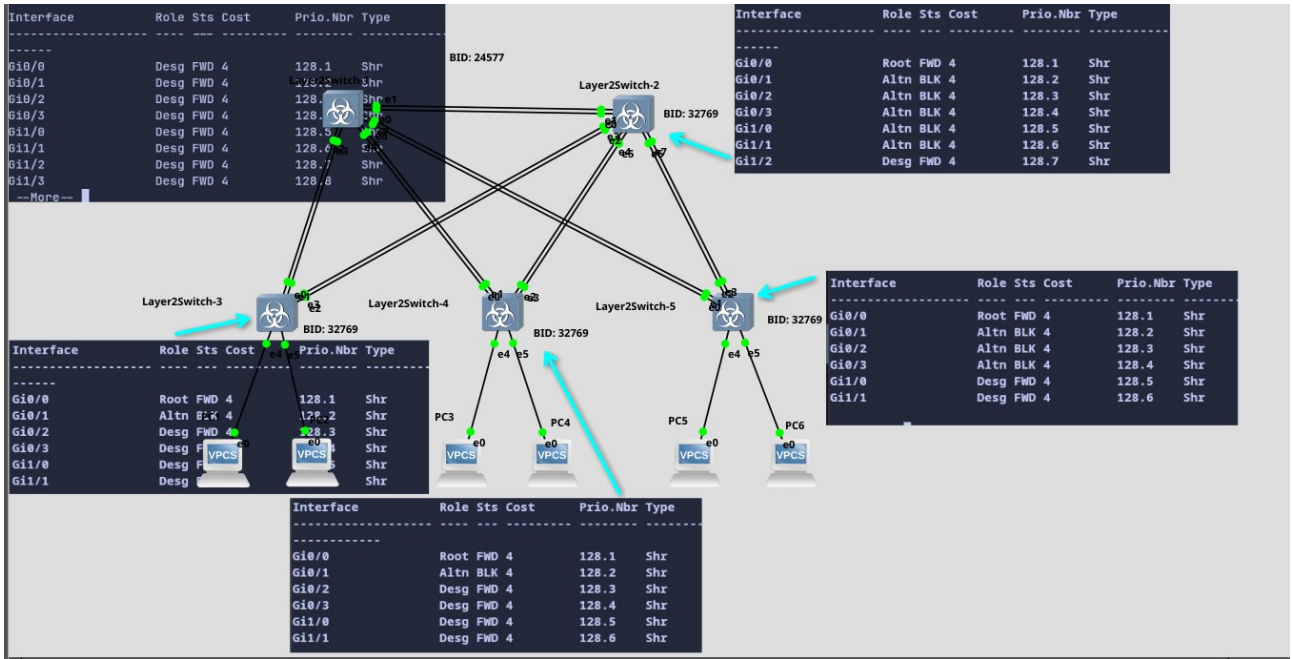
```
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 24577
Address 0c62.e6ee.0000
Cost 4
Port 1 (GigabitEthernet0/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0cf8.f030.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec
```

```
-----
Interface          Role Sts Cost    Prio.Nbr Type
-----
Root FWD 4         128.1  Shr
Altn BLK 4         128.2  Shr
Desg FWD 4         128.3  Shr
Desg FWD 4         128.4  Shr
Desg FWD 4         128.5  Shr
Desg FWD 4         128.6  Shr

G10/0              Root FWD 4         128.1  Shr
G10/1              Altn BLK 4         128.2  Shr
G10/2              Altn BLK 4         128.3  Shr
G10/3              Altn BLK 4         128.4  Shr
G10/0              Altn BLK 4         128.5  Shr
G11/1              Altn BLK 4         128.6  Shr
G11/2              Desg FWD 4         128.7  Shr

--More--
```



4) Каждые 2 секунды hello. Отправляет только root (у меня это switch-1), остальные получают.

The image displays two screenshots of a Wireshark network traffic capture. Both captures are filtered for 'stp' (Spanning Tree Protocol) traffic. The top screenshot shows the first 19 packets, all of which are STP BPDUs (Type 1) sent from the root switch (32768/100/8c:62:e6:ee:00:00) to various other switches. The bottom screenshot shows packets 20 through 14, continuing the sequence of STP BPDUs. The packet list on the right side of each window shows the destination MAC address for each packet, which is the MAC address of the destination switch.

No.	Time	Delta	Source	Destination	Protocol	Length	Info
1	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
2	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
3	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
4	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
5	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
6	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
7	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
8	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
9	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
10	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
11	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
12	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
13	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
14	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
15	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
16	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
17	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
18	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007
19	0.000000		8c:62:e6:ee:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:62:e6:ee:00:00 Cost = 0 Port = 0x8007

No.	Time	Delta	Source	Destination	Protocol	Length	Info
1	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
2	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
3	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
4	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
5	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
6	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
7	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
8	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
9	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
10	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
11	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
12	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
13	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001
14	0.000000		8c:b3:cd:2e:00:00	Nearest-Customer-B...	STP	68	Conf. Root = 32768/100/8c:b3:cd:2e:00:00 Cost = 0 Port = 0x8001

5) Выбрал коммутатор switch-2. Изменил цену к руту на 100.

```
vIOS-L2-01>enable
vIOS-L2-01#
vIOS-L2-01#QQ
vIOS-L2-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#interface gigabitEthernet 0/0
vIOS-L2-01(config-if)#spanning-tree cost 100
      ^
% Invalid input detected at '^' marker.

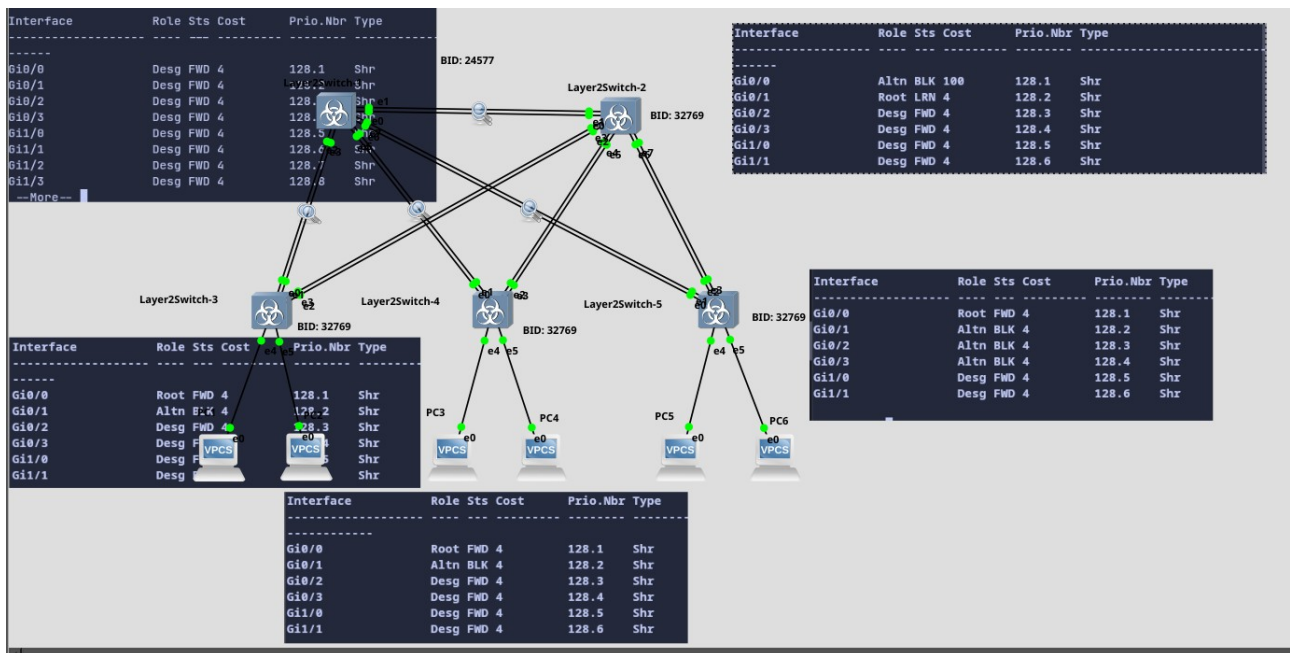
vIOS-L2-01(config-if)#spanning-tree cost 100
vIOS-L2-01(config-if)#end
vIOS-L2-01#
*Jul 29 10:47:36.924: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     0c62.e6ee.0000
             Cost        4
             Port        2 (GigabitEthernet0/1)
             Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0ca2.81bf.0000
             Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
             Aging Time   15 sec

Interface                Role Sts Cost        Prio.Nbr Type
-----
-----
Gi0/0                    Altn BLK 100        128.1    Shr
Gi0/1                    Root LRN 4          128.2    Shr
Gi0/2                    Desg FWD 4          128.3    Shr
Gi0/3                    Desg FWD 4          128.4    Shr
Gi1/0                    Desg FWD 4          128.5    Shr
Gi1/1                    Desg FWD 4          128.6    Shr

--More--
```



6) Команда wr, для записи running-config в startup-config

```

Layer2Switch-1
Translating "wr"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
VIOS-L2-01>show ip
% Incomplete command.

VIOS-L2-01>
VIOS-L2-01>wr
Translating "wr"...domain server (255.255.255.255)
(255.255.255.255)
Translating "wr"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
VIOS-L2-01>
VIOS-L2-01>Q
VIOS-L2-01>Q
VIOS-L2-01>enable
VIOS-L2-01#wr
Building configuration...
Compressed configuration from 5278 bytes to 2019 bytes[OK]
VIOS-L2-01#
*Jul 29 10:27:28.569: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
VIOS-L2-01#
*Jul 29 10:27:29.649: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
VIOS-L2-01#

Layer2Switch-2
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*****
VIOS-L2-01>enable
VIOS-L2-01#wr
Building configuration...
Compressed configuration from 5240 bytes to 1997 bytes[OK]
*Jul 29 10:27:41.006: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jul 29 10:27:42.064: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
VIOS-L2-01#

Layer2Switch-3
* environment.
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*****
VIOS-L2-01#wr
Building configuration...
Compressed configuration from 5866 bytes to 1930 bytes[OK]
*Jul 29 10:54:58.585: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
VIOS-L2-01#
*Jul 29 10:54:59.710: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
VIOS-L2-01#

Layer2Switch-4
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*****
VIOS-L2-01>wr
Translating "wr"...domain server (255.255.255.255)
(255.255.255.255)
Translating "wr"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
VIOS-L2-01>enable
VIOS-L2-01#wr
Building configuration...
Compressed configuration from 5042 bytes to 1921 bytes[OK]
*Jul 29 10:54:02.399: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
VIOS-L2-01#
*Jul 29 10:54:03.601: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
VIOS-L2-01#

Layer2Switch-5
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*
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*****
VIOS-L2-01>enable
VIOS-L2-01#wr
Building configuration...
Compressed configuration from 5042 bytes to 1921 bytes[OK]
*Jul 29 10:56:25.047: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jul 29 10:56:26.134: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
VIOS-L2-01#

```

Для switch нету автоматической записи в конфиг

Config export is not supported by the following nodes: Layer2Switch-1 Layer2Switch-2 Layer2Switch-3 Layer2Switch-4 Layer2Switch-5

Поэтому пишу show running-config и копирую в файлы.

```
> ls
Layer2Switch-1.cfg  Layer2Switch-3.cfg  Layer2Switch-5.cfg  scheme_part5.png
Layer2Switch-2.cfg  Layer2Switch-4.cfg  scheme_part3.png

^> ~/Projects/eltex_summer_course/module4/lab2 > on main !1 ?2
```

Копировал config из каждого switch вручную 😓😓😓